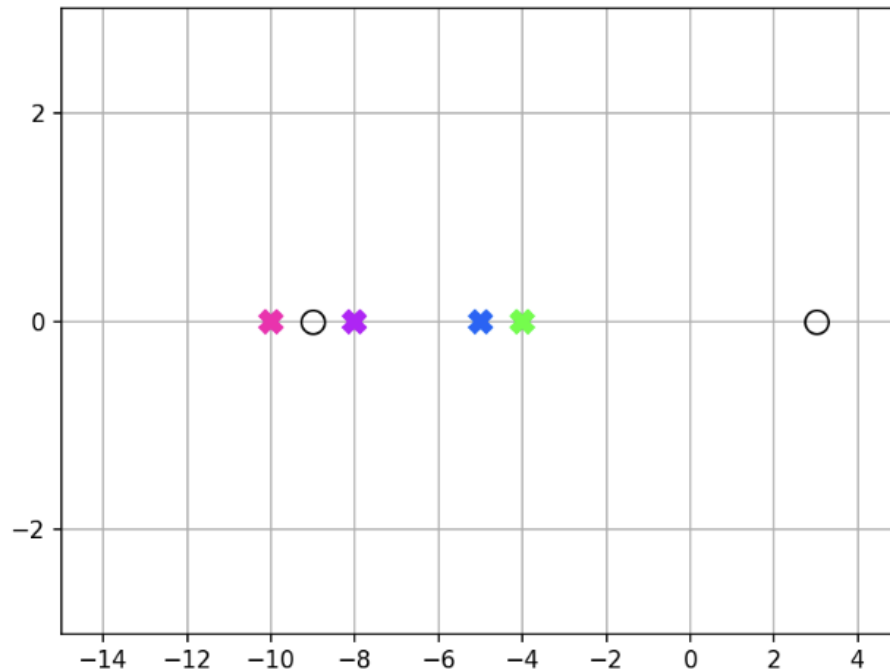


Question 2

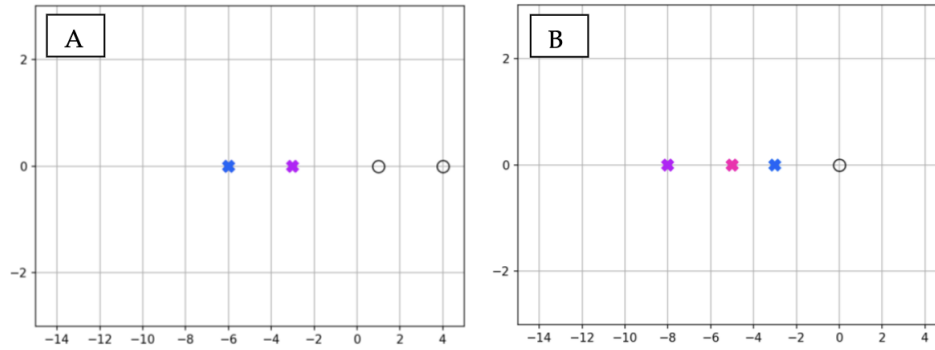
A fourth-order system has poles at -10, -8, -5, and -4, and zeros at -9 and 3:



- Will the system be stable in closed-loop for any gain?
- How many asymptotes will the root locus have? Find their angles and point of intersection on the real axis.
- Sketch a rough root locus plot for the system (you only need to show which sections of the real axis are included and the trajectories of the poles along the asymptotes. Specific values of breakaway points, angles of departure, etc. are not required).
- Based on your plot, answer the following true or false questions about the system (with reasons):
 - A first-order model is an acceptable representation of the system's behaviour at some gains.
 - All the system's modes have a finite maximum frequency.
 - All the system's modes have a finite minimum settling time.
 - Adding a zero at -1 would ensure that the system will be stable for all gains.

Question 3

Consider the pole-zero plots of the following third-order systems (A has a repeating pole at -6):



By sketching rough root locus plots of these systems, determine:

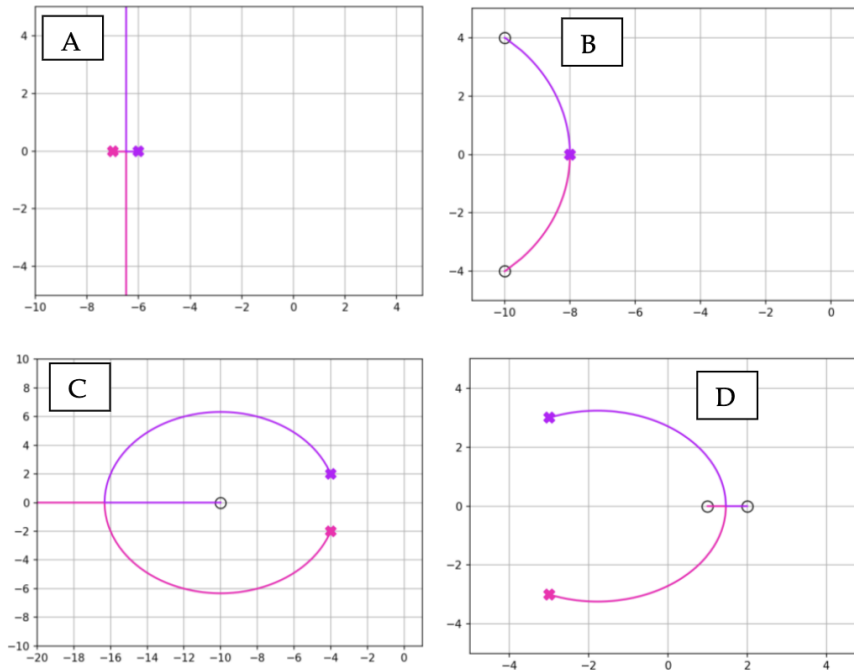
- Which root locus exists on the real axis at -4?
- Which system has a maximum gain?
- Which system's response has a finite maximum frequency?
- Which system can be approximated by a first-order model at some gains?
- Which system can be approximated by a second-order model at some gains?
- Which system's response will settle more quickly as the gain increases?

(answers can be one, or both, or neither)

Bonus: sketch the complete root locus for these systems, including all values of interest.

Question 4

Complete the following root locus sketches by filling in all values of interest (asymptotes, breakaway/-in points, imaginary axis crossings, angles of departure/arrival). The real and imaginary parts of all poles and zeros are integers.



Question 5

A spring-mass-damper system has a mass $m = 1$, damping coefficient $b = 3$ and spring constant $k = 6$. The transfer function of the system is

$$\frac{1}{ms^2 + bs + k}$$

- Sketch a root locus showing how variation in the i) mass ii) damping coefficient and iii) spring constant affects the pole positions. (Keep the other parameters constant at their nominal values in each case).
- If each of these parameters is subject to a 5% tolerance, calculate the (maximum) root sensitivity within this range of the nominal value. Which parameter's fluctuations will affect the system's response the most?